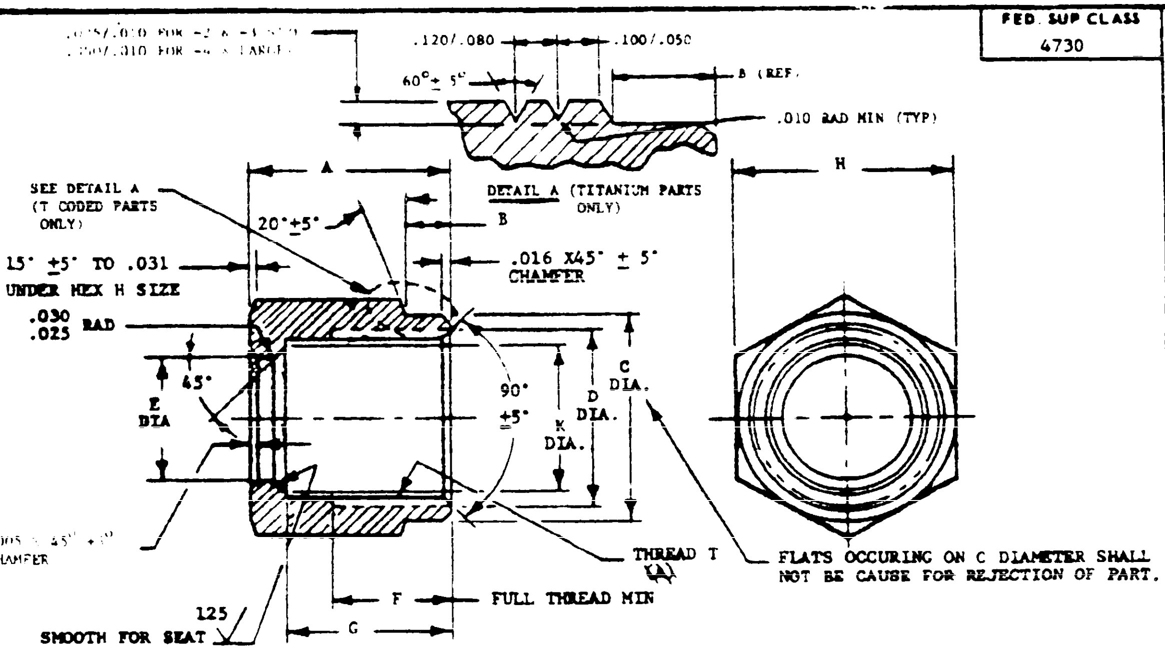
USER SYMBOLS:
NAVY - OS
ARMY - CI, CR, MI, AT

REVIEWER SYMBOLS:
ARMY - WC, MI

"Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS."

FED. SUP CLASS
4730


.025/.010 FOR -2 & -3 SIZE
.050/.010 FOR -4 & LARGER
.120/.080
.100/.050
60° ± 5°
B (REF)
.010 RAD MIN (TYP)
DETAIL A (TITANIUM PARTS ONLY)
B
H
A
SEE DETAIL A (T CODED PARTS ONLY)
20° ±5°
15° ±5° TO .031 UNDER HEX H SIZE
.016 X45° ± 5° CHAMFER
.030 .025 RAD
45°
E DIA
90° ±5°
C DIA.
D DIA.
K DIA.
.005 X 45° ±5° CHAMFER
THREAD T (A)
FLATS OCCURING ON C DIAMETER SHALL NOT BE CAUSE FOR REJECTION OF PART.
F
FULL THREAD MIN
G
125
SMOOTH FOR SEAT

~~(a) PARTS WITH THREADS IN ACCORDANCE WITH MIL-S-7742 OF THE SAME THREAD SIZE, PITCH AND TOLERANCE CLASS ARE ACCEPTABLE FOR PROCUREMENT UNTIL 1 JULY 1971, BUT PARTS WITH MIL-S-7742 THREADS WILL NOT BE PROCURED THEREAFTER. EXISTING STOCKS ARE TO BE USED UNTIL EXHAUSTED.~~

(b) USE TAP DRILL DIA FOR K EXCEPT IN SIZES 20, 24 AND 32.

(c) REVISION "F" SIZE 20 PARTS WITH G DIMENSION 0.687 AND K DIMENSION 1.456; AND SIZE 24 PARTS WITH G DIMENSION 0.687 AND K DIMENSION 1.705; ARE INACTIVE FOR DESIGN AND ARE NOT ACCEPTABLE FOR PROCUREMENT AFTER 14 JULY 1975. EXISTING STOCKS ARE TO BE USED UNTIL EXHAUSTED.

NOTES:

1. MATERIAL: CARBON STEEL: BARS, RODS, PLATES AND FORGINGS
   ALUMINUM ALLOY: BARS, RODS AND FORGINGS
   CORROSION RESISTANT STEEL: BARS AND FORGINGS
   Ⓗ TITANIUM ALLOY - BARS

   ⓒ SEE PROCUREMENT SPECIFICATION
   NOTE: PARTS MADE FROM 1137 OR 1141 FORGINGS QUALIFIED TO MIL-F-18290 SHALL NOT BE SUPPLIED AFTER 30 JUNE 1984.
2. FINISH: SEE PROCUREMENT SPECIFICATION.
3. SURFACE TEXTURE: IN ACCORDANCE WITH ANSI B46.1.
4. BREAK ALL SHARP EDGES AND REMOVE ALL HANGING BURRS AND SLIVERS.
5. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.005, ANGLES ±1/2°.
6. CIRCULAR SURFACES OF DIAMETERS E, K, AND THREAD T SHALL BE CONCENTRIC WITHIN .005 TIR.
7. INTERCHANGEABILITY RELATIONSHIP: MS21917 AND MS21921 PARTS IDENTIFIED BY THE SAME DASH NUMBERS ARE FUNCTIONALLY INTERCHANGEABLE. USE MS21921 AND MS21922 AS AN ASSEMBLY WHEN SUBSTITUTING FOR MS21917 AND MS21918.
8. Ⓗ TITANIUM NOT RECOMMENDED FOR USE IN OXYGEN SYSTEMS.

* CERTAIN PROVISIONS (SCREW THREADS AND HEXAGON SIZE) OF THIS STANDARD ARE THE SUBJECT OF INTERNATIONAL STANDARDIZATION AGREEMENT ASCC AIR STD 17/15. WHEN REVISION, OR CANCELLATION OF THIS STANDARD IS PROPOSED WHICH WILL AFFECT OR VIOLATE THE INTERNATIONAL AGREEMENT CONCERNED, THE PREPARING ACTIVITY WILL TAKE APPROPRIATE RECONCILIATION ACTION THROUGH INTERNATIONAL STANDARDIZATION CHANNELS, INCLUDING DEPARTMENTAL STANDARDIZATION OFFICES, IF REQUIRED.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

Ⓗ DENOTES CHANGES

APPROVED 27 FEB 56 REVISED Ⓕ 2 MAR 73 Ⓖ 14 JULY 1977 Ⓗ 26 JUNE 1981

"This military standard is approved for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document"

P.A. NAVY - AS Other Cust USAF-99 ARMY-AV	INTERNATIONAL INTEREST *	TITLE NUT, SLEEVE COUPLING, FLARELESS	MILITARY STANDARD MS21921
PROCUREMENT SPECIFICATION MIL-F-18280	SUPERSEDES MS 21917		SHEET 1 OF 2

DD FORM 1 MAR 77 672-1 (Coordinated)

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PLATE NO. 23088

PROJECT NO. 4730-0161

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FED. SUP CLASS
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(H)

DASH NO.	~~DASH NO.~~ ALUMINUM ALLOY		CRES CLASS 304	316	347	INACTIVE	TUBE O.D.	THREAD T (4) MIL-S-8879	A ±.010	B ±.010	C +.005 -.000	D ±.016	E +.005 -.000
- 2	-2D	-2W	-2J	-2K	-2S	~~-2C~~	1/8	[illegible]	.53	.14	.365	.328	.171
- 3	-3D	-3W	-3J	-3K	-3S	~~-3C~~	3/16	.3750-24 UNJF-3B	.58	.10	.427	.391	.232
- 4	-4D	-4W	-4J	-4K	-4S	~~-4C~~	1/4	.4375-20 UNJF-3B	.72	.18	.552	.453	.300
- 5	-5D	-5W	-5J	-5K	-5S	~~-5C~~	5/16	.5000-20 UNJF-3B	.76		.615	.516	.366
- 6	-6D	-6W	-6J	-6K	-6S	~~-6C~~	3/8	.5625-18 UNJF-3B	.80		.677	.578	.425
- 8	-8D	-8W	-8J	-8K	-8S	~~-8C~~	1/2	.7500-16 UNJF-3B	.86		.865	.766	.552
-10	-10D	-10W	-10J	-10K	-10S	~~-10C~~	5/8	.8750-14 UNJF-3B	.98	.22	.990	.891	.690
-12	-12D	-12W	-12J	-12K	-12S	~~-12C~~	3/4	1.0625-12 UNJ -3B	1.00	.24	1.240	1.078	.826
-16	-16D	-16W	-16J	-16K	-16S	~~-16C~~	1	1.3125-12 UNJ -3B	1.05	.26	1.490	1.328	1.074
-20	-20D	-20W	-20J	-20K	-20S	~~-20C~~	1-1/4	1.6250-12 UNJ -3B			1.990	1.641	1.330
-24	-24D	-24W	-24J	-24K	-24S	~~-24C~~	1-1/2	1.8750-12 UNJ -3B	1.03	.22	2.236	1.891	1.580
-32	-32D	-32W	-32J	-32K	-32S	~~-32C~~	2	2.5000-12 UNJ -3B	1.12		2.365	2.562	2.080

(H)

DASH NO.	ALUMINUM ALLOY		CRES CLASS 304	316	347	INACTIVE	MIN.	(c) G ±.010	H		(b)(c) K ±.010	WEIGHT LB MAX STEEL	AL ALLOY	TI ALLOY
-2	- 2D	-2W	-2J	-2K	-2S	~~-2C~~	.360	.425	.375	+.003 -.004		.008	.003	.004
-3	- 3D	-3W	-3J	-3K	-3S	~~-3C~~	.366	.429	.438			.010	.004	.005
-4	- 4D	-4W	-4J	-4K	-4S	~~-4C~~	.431	.547	.562			.028	.010	.016
-5	- 5D	-5W	-5J	-5K	-5S	~~-5C~~	.444	.562	.625			.032	.012	.018
-6	- 6D	-6W	-6J	-6K	-6S	~~-6C~~	.422	.578	.688			.037	.014	.021
-8	- 8D	-8W	-8J	-8K	-8S	~~-8C~~	.516	.609	.875			.070	.025	.040
-10	-10D	-10W	-10J	-10K	-10S	~~-10C~~	.578	.687	1.000			.083	.031	.047
-12	-12D	-12W	-12J	-12K	-12S	~~-12C~~	.562		1.250			.148	.054	.085
-16	-16D	-16W	-16J	-16K	-16S	~~-16C~~			1.500			.220	.078	.126
-20	-20D	-20W	-20J	-20K	-20S	~~-20C~~		.675	2.000	±.016	1.482	.470	.181	.271
-24	-24D	-24W	-24J	-24K	-24S	~~-24C~~			2.250	±.020	1.731	.530	250	.305
-32	-32D	-32W	-32J	-32K	-32S	~~-32C~~		.587	2.375		2.295	.360	.413	.496

(H) ITEMS CODED "C" ON PREVIOUS ISSUES OF THIS STANDARD ARE NO LONGER TO BE FURNISHED. UNLESS OTHERWISE SPECIFIED, PARTS CODED J, K, OR S (AT SUPPLIER'S OPTION) SHALL BE FURNISHED IN LIEU OF "C" CODED PARTS.

ADD D AFTER DASH NO. FOR ALUMINUM ALLOY (2014) (2024).
ADD W AFTER DASH NO. FOR ALUMINUM ALLOY (7075).
ADD J AFTER DASH NO. FOR CORROSION RESISTANT STEEL, CLASS 304.
ADD K AFTER DASH NO. FOR CORROSION RESISTANT STEEL, CLASS 316.
ADD S AFTER DASH NO. FOR CORROSION RESISTANT STEEL, CLASS 347.
(H) ADD T AFTER DASH NO. FOR TITANIUM ALLOY (6AL-4V).

EXAMPLE OF PART NUMBERS:
MS21921-4 - NUT, 1/4 TUBING, CARBON STEEL.
MS21921-8D - NUT, 1/2 TUBING, ALUMINUM ALLOY (2014) (2024).
MS21921-8W - NUT, 1/2 TUBING, ALUMINUM ALLOY (7075).
MS21921-8J - NUT, 1/2 TUBING, CORROSION RESISTANT STEEL, CLASS 304.
MS21921-8K - NUT, 1/2 TUBING, CORROSION RESISTANT STEEL, CLASS 316.
MS21921-8S - NUT, 1/2 TUBING, CORROSION RESISTANT STEEL, CLASS 347.
(H) MS21921-8T - NUT, 1/2 TUBING, TITANIUM ALLOY (6AL-4V).

APPROVED 27 FEB 56 REVISED (H)

P.A. NAVY - AS Other Cust USAF - 99 ARMY-AV	INTERNATIONAL INTEREST *	TITLE NUT, SLEEVE COUPLING, FLARELESS	MILITARY STANDARD MS21921
PROCUREMENT SPECIFICATION MIL-F-18280	SUPERSEDES MS 21917		SHEET 2 OF 2

DD FORM MAR 57 672-1 (Continued)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

USER SYMBOLS
NAVY - AS
ARMY - AV

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